CLOUDSHIELD TECHNOLOGIES

Backup Restoration

Control Test Report

GRC-004: Inadequate Backup Restoration Testing

Independent GRC Portfolio Project: Simulated Organization

Test ID	BRC-001
System	PostgreSQL Customer Database
Control Owner	Infrastructure / Business Continuity
Test Type	Backup Restoration / Recovery Control Test
Overall Result	PASS WITH OBSERVATION / NEEDS IMPROVEMENT
Linked Risk	GRC-004
Risk Treatment	Mitigate
Retest Required	Yes: next quarterly restoration test

1. Executive Summary

CloudShield tested restoration of the PostgreSQL customer database to determine whether backup and recovery controls operate effectively and whether recovery objectives can be achieved. The restoration completed successfully within both the Recovery Time Objective (RTO) and Recovery Point Objective (RPO). However, the documented recovery runbook contained an incorrect command, and successful completion depended on the engineer's knowledge to work around the error.

Overall result: PASS WITH OBSERVATION / NEEDS IMPROVEMENT. The recovery capability worked, but the documentation weakness should be corrected and formally retested.

2. Control Objective

Verify that CloudShield can restore the customer database from backup within approved recovery objectives and that the recovery process is sufficiently documented, repeatable, and supported by retained evidence.

3. Recovery Objectives

Recovery Requirement	CloudShield Target
Recovery Time Objective (RTO)	4 hours: maximum target time to restore the database to an operational state.
Recovery Point Objective (RPO)	24 hours: maximum acceptable age of recoverable data / potential data loss window.
Testing Expectation	Documented restoration testing should occur quarterly, with results and corrective actions retained.

4. Evidence Reviewed

- Daily backup job/log showing the selected backup completed successfully.
- Backup timestamp demonstrating the recovery point used for the test.
- Restore-test ticket / test record.
- Start and completion timestamps for the restoration.

- System evidence confirming the restored database became operational in the isolated test environment.
- Existing database recovery runbook.
- Prior restoration-test records, showing no documented restoration test during the preceding 11 months.

5. Test Procedure

1. Confirm the most recent eligible database backup completed successfully and record its timestamp.

2. Create or verify an isolated test environment suitable for restoration.

3. Record the restoration start time.

4. Follow the documented recovery runbook to restore the PostgreSQL customer database.

5. Record deviations, errors, or manual workarounds encountered during the procedure.

6. Verify the restored database becomes operational and record the completion time.

7. Calculate actual recovery duration and compare it with the 4-hour RTO.

8. Calculate the age of the restored backup and compare it with the 24-hour RPO.

9. Document the result, observation, remediation owner, and retest requirement.

6. Test Results

Measure	**Target**	**Observed Result**	**Assessment**	**Evidence / Calculation**
Backup availability	Successful backup available	Backup available and selected	PASS	Daily backup log / selected restore point
RTO	≤ 4 hours	2 hours 18 minutes	PASS	09:00 start to 11:18 operational
RPO	≤ 24 hours	6 hours	PASS	Selected backup was 6 hours old
Runbook accuracy	Procedure should be accurate and	One documented command was	NEEDS IMPROVEMENT	Engineer used technical

Measure	Target	Observed Result	Assessment	Evidence / Calculation
	repeatable	incorrect		knowledge to complete restoration
Testing cadence	Quarterly documented restoration test	No documented test for 11 months	NEEDS IMPROVEMENT	Prior test evidence / control history

7. RTO and RPO Analysis

Measure	Analysis
RTO calculation	11:18 AM completion − 9:00 AM start = 2 hours 18 minutes.
RTO conclusion	PASS. Actual recovery time of 2h 18m is 1h 42m inside the 4-hour target.
RPO calculation	The restored backup was 6 hours old at the time of recovery testing.
RPO conclusion	PASS. A 6-hour recovery point is 18 hours inside the maximum 24-hour data-loss window.

8. Control Observation

The recovery process is technically capable of restoring the database within approved RTO and RPO targets. However, the documented recovery runbook is not fully reliable because one command was incorrect. The engineer recognized the problem and completed the restoration using technical knowledge not captured in the documented procedure.

This creates key-person dependency: a future recovery attempt performed by a different responder could take longer, fail, or exceed the RTO if the same documentation defect is encountered.

9. Root Cause & Risk

Assessment	Conclusion
Root Cause	Recovery documentation was not validated through sufficiently frequent formal restoration testing.
Risk	Outdated or inaccurate instructions may delay restoration during a real disruption, increase dependence on individual staff knowledge, and reduce confidence that recovery is repeatable.
Risk Rating	Medium — the tested recovery capability worked, but process reliability and evidence need improvement.
Control Conclusion	Design is generally adequate; operating effectiveness needs improvement because testing/documentation discipline was insufficient.

10. Corrective Action Plan

Action ID	Corrective Action	Owner	Target	Status
BRC-CAR-01	Correct the inaccurate command in the database recovery runbook and perform peer review.	Infrastructure / DBA	Within 10 business days	Open
BRC-CAR-02	Establish and document quarterly backup	Infrastructure / Business	Before next quarter	Open

Action ID	Corrective Action	Owner	Target	Status
	restoration testing.	Continuity		
BRC-CAR-03	Retain test evidence including timestamps, backup age, RTO/RPO results, issues, and corrective actions.	Infrastructure / Security	Ongoing	Open
BRC-CAR-04	Retest using the corrected runbook with a responder who did not rely on undocumented knowledge.	Infrastructure / Security	Next quarterly test	Open

11. Management Response

Management accepts the observation and agrees to update the recovery runbook, establish quarterly restoration testing, retain evidence, and verify the revised procedure during the next scheduled test. The finding remains open until corrective actions are completed and the updated process is successfully retested.

12. Retest Plan

- Use the corrected recovery runbook without relying on undocumented steps.
- Have a qualified responder other than the original engineer execute or independently validate the procedure.
- Confirm successful restoration and record start/completion timestamps.
- Confirm actual RTO remains within four hours.
- Confirm the selected recovery point remains within the 24-hour RPO.

- Retain screenshots/logs, test record, issues, approvals, and closure evidence.
- Close GRC-004 only after the corrective actions and retest evidence demonstrate a repeatable recovery process.

13. Final Assessment

PASS WITH OBSERVATION / NEEDS IMPROVEMENT. CloudShield demonstrated that the PostgreSQL customer database can be restored within established RTO and RPO targets. The control remains subject to remediation because the recovery documentation was inaccurate and formal restoration testing had not been performed for 11 months.

Appendix A — Control Test Evidence Record

Evidence Field	Recorded Result
Test ID	BRC-001
Linked Risk	GRC-004
System	PostgreSQL Customer Database
Restore Start	9:00 AM
Operational / Test Complete	11:18 AM
Actual RTO	2 hours 18 minutes
RTO Target	4 hours
Backup Age / Actual RPO	6 hours
RPO Target	24 hours
Procedure Exception	Incorrect runbook command; engineer used technical knowledge to continue.
Overall Result	Pass with Observation / Needs Improvement
Retest	Required: next quarterly restoration test